[journal]IEEEtrandocumentabstractThis paper presents AER-MQoS, a context-aware multi-objective extension of the RPL routing protocol for low-power and lossy IoT networks. AER-MQoS unifies traffic-class-aware Quality of Service (QoS) differentiation, energy awareness, and link-reliability-informed parent selection through a single Multi-Criteria Score (MCS) driven by a bounded context coefficient γ that dynamically balances QoS responsiveness (α) and energy preservation (β) according to both application domain and residual energy. Weighted round-robin application queues (6:3:2:1) enforce class-based scheduling at the UDP layer, mitigating head-of-line blocking without modifying the RPL control plane. The protocol is implemented as an experimental Objective Code Point (OCP 8) on Contiki-NG and evaluated through reproducible Cooja simulations (N=4 seeds, 1800 s, 25-node topology) across four firmware variants: RPL_STANDARD, RPL_MQOS, RPL_AER, and the integrated AER-MQoS. Under lossy UDGM channels ($success\_ratio$=0.85), all four variants achieve $> 99.9\%$ PDR with overlapping descriptive ranges, confirming correct four-way integration and establishing a baseline for planned stress-test campaigns. The primary contributions are: (i) a unified contextual model (α, β, γ) that makes the QoS–energy trade-off explicit and measurable; (ii) a composite path decision rule (MCS) integrated with Minimum Rank with Hysteresis Objective Function (MRHOF)-compatible hysteresis; (iii) a lightweight WRR queuing layer operating above UDP for portable class isolation; and (iv) an open-source Contiki-NG implementation with reproducible CSV-to-figure provenance. Energest-calibrated energy validation, attack-injection scenarios, and higher-stress confirmatory campaigns are identified as the next milestones.